Paleomagnetism of Eocene rocks of the Kelowna and Castlegar areas, British Columbia: studies in determining paleohorizontal[1]

M. BARDOUX[2]
Ottawa–Carleton Centre for Geoscience Studies, Carleton University, Ottawa, Ont., Canada K1S 5B6
AND
E. IRVING
Pacific Geoscience Centre, Geological Survey of Canada, P.O. Box 6000, Sidney, B.C., Canada V8L 4B2

Received June 7, 1988
Revision accepted November 22, 1988

The middle Eocene Marron volcanics (mean age 52 ± 2 Ma) of the Kelowna outlier form the upper part of the hanging wall of the westerly dipping Okanagan Valley fault (OVF) in south-central British Columbia. They overlie Quesnellia. The OVF is currently interpreted as the westernmost member of a network of low-angle extension faults in the southern Omenica belt. The OVF was active in the middle Eocene at much the same time that the Marron volcanics were cooling. Relative to present horizontal, the magnetizations are widely scattered (Fisher's precision parameter $k = 8$) and after correction for bedding attitudes, there is no significant improvement ($k = 9$). Evidently, some magnetizations were acquired before (referred to as category 1) and others after (category 2) tilting; that is, the horizontal plane at the time of magnetization sometimes did and sometimes did not coincide with the bedding plane. Partial unfolding experiments, carried out on the two categories separately, yield a precision comparable to that expected for paleosecular variation, and a mean direction (D, I) of 352°, 70° (24 sites spanning 2000 m, 275 specimens, $k = 23$, $\alpha_{95} = 6°$, paleopole 86°N, 230°E, $A_{95} = 10°$). The Marron is predominantly normally magnetized. Rock units slightly older and others slightly younger are reversely magnetized. The transition from reversed to normal polarity occurs in basal beds of the Marron Formation. The overall mean direction of the Marron and stratigraphically adjacent units is 352°, 69° (28 sites, 300 specimens, spaning 4000 m, $k = 21$, $\alpha_{95} = 6°$), yielding a paleopole at 85°N, 197°E ($A_{95} = 10°$), which is in excellent agreement with that for middle Eocene rocks of cratonic North America. Hence this part of Quesnellia had reached its present position relative to North America by middle Eocene time, and there has been no significant rotation of it. In contrast, the mean direction (020°, 72°, $k = 9$, $\alpha_{95} = 11°$) after correction for bedding (calculated assuming the magnetization to be *entirely* pretilting) implies a clockwise rotation of 28°. We believe that this is incorrect; the apparent rotation, we argue, is caused by wrongly assuming that the bedding plane always coincides with the paleohorizontal at the time magnetism is acquired.

Further tests have been carried out on intrusive and metamorphic core-complex rocks in the region of Eocene crustal extension 100 km to the east of Kelowna. These rocks are coeval with the Marron, and are located in both the hanging walls and footwalls of the Slocan Lake normal extension fault, which dips 30° eastward. Paleodirections are very different from those at Kelowna (four bodies, mean direction (D, I) 60°, 52°, $k = 66$, $\alpha_{95} = 6°$), and we argue that this divergence is caused by tilting 37 ± 10° to the west antithetical to the Slocan Lake fault. We suggest that paleomagnetism provides a means by which tilts in such plutonic and metamorphic terrains can be determined. We suggest further that such tilts may have been responsible for some of the aberrant magnetizations observed in plutonic rocks of the Coast Plutonic Complex being much more widespread in the cordillera than previously envisioned.

Les volcanites de Marron, d'âge éocène moyen (moyenne = 52 ± 2 Ma), du lambeau d'érosion Kelowna forment la partie élevée de la lèvre supérieure de la faille de la vallée d'Okanagan, inclinée vers l'ouest, dans le centre-sud de la Colombie-Britannique. Elles reposent sur les assises du Quesnellia. La faille de la vallée d'Okanagan est couramment interprétée comme la plus occidentale des failles distensives subhorizontales d'un réseau localisé dans la ceinture d'Omenica sud. La faille de la vallée d'Okanagan était active à l'Éocène moyen, en bonne partie durant le refroidissement des volcanites de Marron. En regard de l'horizontalité actuelle, les aimantations sont très fortement dispersées (paramètre de précision de Fisher $k = 8$) et une correction pour les attitudes du litage n'a fourni aucune amélioration significative ($k = 9$). Il est évident que certaines aimantations furent acquises avant le basculement (désignées catégorie 1), tandis que pour d'autres elles le furent après (désignées catégorie 2); ce qui signifie que le plan horizontal au moment de l'aimantation pouvait quelques fois coïncider et en d'autres fois ne pas coïncider avec le plan de stratification. Des expériences de déplissement partiel, réalisées séparément sur les deux catégories, fournissent une précision comparable à celle attendue d'une variation paléoséculaire, et une direction moyenne (D, I) de 352°, 70° (24 sites, étalement 2000 m, 275 spécimens, $k = 23$, $\alpha_{95} = 6°$, paléopôle 86°N, 230°E, $A_{95} = 10°$). En général, les volcanites de Marron possèdent une aimantation normale. Les unités lithologiques légèrement plus âgées ainsi que les autres légèrement plus jeunes sont aimantées inversement. La transition entre la polarité inverse et normale apparaît dans les couches basales de la Formation de Marron. La direction globale moyenne de la Formation de Marron et des unités stratigraphiques adjacentes est 352°, 69° (28 sites, 300 spécimens, étalement 4000 m, $k = 21$, $\alpha_{95} = 6°$), fournissant un paléopôle à 85°N, 197°E ($A_{95} = 10°$), laquelle direction correspond de près à celle des roches de l'Éocène moyen du craton de l'Amérique du Nord. En conséquence cette portion du Quesnellia aurait atteint sa position actuelle par rapport à l'Amérique du Nord au temps de l'Éocène moyen, sans avoir été affectée par une rotation

Can. J. Earth Sci. **26**, 829–844 (1989)

[1]Geological Survey of Canada Contribution 38188.

[2]Present address: Départment des sciences de la Terre, Université du Québec à Montréal, P.O. Box 8888, Station A, Montréal, Que., Canada H3C 3P8.

Printed in Canada / Imprimé au Canada

significative. Au contraire, la direction moyenne (020°, 72°, $k = 9$, $\alpha_{95} = 11°$) après correction pour le litage (calculée en présumant que l'aimantation fut intégralement acquise avant le basculement) implique une rotation de 28° dans le sens des aiguilles d'une montre. Nous croyons cette interprétation fausse; la rotation apparente serait due à la présomption erronnée que le plan du litage coïncidait toujours avec la paléohorizontalité au temps de l'acquisition de l'aimantation.

Des essais supplémentaires ont été complétés sur des carottes d'un complexe de roches métamorphiques dans la région de distension crustale éocène, à 100 km à l'est de Kelowna. Ces roches sont contemporaines des volcanites de Marron, et elles forment les lèvres supérieure et inférieure de la faille normale distensive du lac Slocan, lacquelle est inclinée à 30° vers l'est. Les paléodirections sont différentes de celles déterminées à Kelowna (quatre corps, direction moyenne (D, I) 60°, 52°, $k = 66$, $\alpha_{95} = 6°$), et il apparaît que cette divergence est due à un basculement de 37 ± 10° à l'ouest, antithétique à la faille du lac Slocan. Il est suggéré que le paléomagnétisme peut fournir un moyen de déterminer les basculements dans de tels terrains plutoniques et métamorphiques. En plus, il est proposé que de tels basculements aient pu être responsables de certaines des aimentations aberrantes observées dans les roches plutoniques du Complexe plutonique côtier étant bien plus répandues dans la cordillère qu'on l'envisageait antérieurement.

[Traduit par la revue]

Introduction

Success in paleomagnetic work depends critically on one being able to determine the attitude of the magnetization horizontal plane. The magnetization horizontal plane (or more briefly the paleomagnetic horizontal) is the horizontal plane at the time the rocks were magnetized. In this paper we describe studies of coeval upper and middle crustal rocks of middle Eocene age from southeastern British Columbia that were made to investigate this problem.

In bedding sequences with a primary magnetization, the geologically determined bedding plane is a good approximation to the paleomagnetic horizontal. In bedded sequences whose magnetization is entirely secondary, the present horizontal usually is assumed to approximate the paleomagnetic horizontal because the dispersion of paleodirections is markedly better before than after correction for bedding attitudes. This simple—either/or—rationale, while still applicable to rock units with uncomplicated histories, is now known to be inapplicable in many instances because magnetization processes can occur over an interval of time that extends from before to after deformation. As the list of rock units with fairly simple thermal and chemical histories is exhausted, and those with more complex histories come to be studied, paleomagnetists will encounter more and more frequently such extended magnetization processes, and will have to face the concomitant problem of determining the paleomagnetic horizontal.

One of the hallmarks of rock units with such extended magnetization histories is that the dispersion of paleodirections is high both before and after correction for bedding—higher, that is, than can be reasonably anticipated from paleosecular variation. Certain studies have shown that partial correction for bedding attitudes produces acceptable dispersions (Schwartz and van der Voo 1984), as if magnetizations were acquired at a certain stage during the deformation process, perhaps even controlled by it. This is unlikely to be generally true because many different combinations of primary, syndeformational, and postdeformational magnetization may be expected to occur within single rock units. For example, one recent study has revealed magnetizations acquired before and during deformation, evidently by two quite different processes, because the two magnetizations had different unblocking temperatures (Miller and Kent 1986).

Rocks may be expected to acquire their magnetization at deposition or during diagenesis by a variety of thermal and chemical processes, and it is the variety of these processes and their temporal relationship to deformation that creates the difficulty in determining the paleomagnetic horizontal. In the first part of this paper we describe the magnetization of a volcanic sequence in the vicinity of Kelowna, British Columbia. The rocks have been variably tilted, and have both pre- and post-tilting magnetizations. The scalar properties of these magnetizations are indistinguishable, and they must, therefore, be separated by their directional characteristics. A rationale for dealing with this situation is presented and its tectonic consequences discussed.

The second topic in this paper concerns the paleomagnetism of rocks that originated in the middle crust. At first sight, observations from such deep-seated rocks may appear difficult to interpret, but, somewhat paradoxically, the problem of determining the paleomagnetic horizontal may be less severe here than in their upper crustal, bedded counterparts. This is because middle (and lower) crustal rocks acquire a magnetization thermally through slow uplift and cooling; their magnetizations are acquired by a relatively simple regional process rather than by variable local processes of deposition and diagenesis. In this situation, the paleomagnetic horizontal is the horizontal plane within the crust as it uplifts and cools. Hence, if the paleofield is known for the period of time in question, then tilts that have occurred after original uplift will be evidenced by departures of the paleomagnetic directions from those expected. From the departures, tilts can be calculated. In the second part of this paper we describe data from metamorphic core-complexes near Castlegar. The data, although few, are introduced in order to suggest a means of estimating crustal tilts in plutonic and high-grade metamorphic rocks.

Geology and sampling

The Eocene volcanics of southeast British Columbia are an erosional remnant of a once continuous volcanic belt extending from Kamloops in British Columbia (Ewing 1981) south to Republic in Washington (Heller *et al.* 1987) and possibly beyond. Total accumulation is estimated to be about 2×10^4 km^3, assuming a length of 300 km, a width of 30 km, and an average thickness of 2 km.

The rocks studied are mainly from the middle Eocene (52 Ma) Marron Formation of south-central British Columbia (Church 1973, 1982*b*; Little 1961, 1983; Monger 1968). They occur in the hanging wall of the Okanagan Valley fault (OVF) near Kelowna. The OVF (Figs. 1–3) marks the boundary between the Omineca (high-grade) and Intermontane (low-grade) belts (Ross 1981; Monger *et al.* 1982). It is 350 km in length and is one of several very large low-angle extension faults that occur in the region of Eocene extension in southeastern British Columbia (Parrish and Carr 1986; Haugerud 1986; Tempelman-Kluit and Parkinson 1986; Brown and Journeay 1987; Parrish *et al.* 1988).

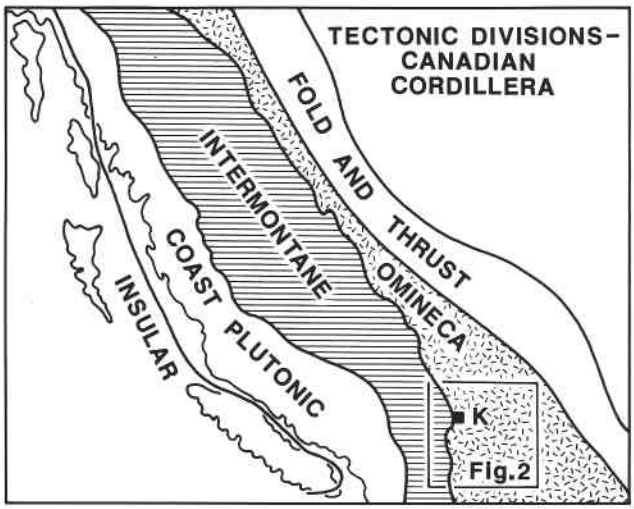


FIG. 1. Relation of sampling region of Fig. 2 to major tectonic belts of the Canadian Cordillera. K, Kelowna.

Mineral-stretching lineations in footwall gneisses adjacent to the OVF are parallel, and trending consistently to the north-northwest, averaging 285° east of north. Mylonites associated with the OVF contain kinematic indicators that suggest a westerly sense of motion of the hanging wall relative to the footwall. U−Pb dates on zircons and monazites from both westerly and easterly dipping faults indicate that ductile deformation associated with motions on them occurred between 58 and 45 Ma (Parrish *et al.* 1988). Some later motion may have occurred on brittle faults that cut the mylonite zones. One of the major faults, the Slocan Lake fault (Carr 1986), has been traced into the lower crust as an easterly dipping seismic reflector (Cook *et al.* 1988). Displacements on the main faults may have been several tens of kilometres (Carr 1985; Bardoux 1986; Tempelman-Kluit and Parkinson 1986; Parrish *et al.* 1988).

Footwall rocks are typically amphibolite-grade gneiss, into which have been intruded bodies of Eocene age—the Coryell syenites—that are essentially coeval with the volcanics of the Marron Formation (Table 1) and may be feeders to them (Tempelman-Kluit and Parkinson 1986). Above the OVF, the hanging wall is composed of low-grade late Paleozoic and Mesozoic eugeosynclincal rocks cut by Jurassic plutons. K−Ar mica dates of Jurassic age (Medford 1975) are preserved in the plutons, indicating that they, and the hanging-wall rocks, have not been heated above 300°C since emplacement. In contrast, K−Ar dates from micas and hornblende and fission-track dates on apatite and sphene from footwall gneisses of the OVF indicate that they were heated to a temperature above 500°C until 52 Ma, and then cooled rapidly to 100°C within about 5 Ma (Medford 1975; Parkinson 1985; Mathews 1981). Such metamorphic rocks might be expected to preserve a record of the middle Eocene geomagnetic field.

The sampling region in the vicinity of Kelowna is shown in Fig. 3. The volcanics are accessible by road and are exposed in long semicontinuous sections, capped by late middle Eocene lacustrine sedimentary rocks, which provide excellent determinations of bedding attitude. The rareness of faults cutting the sequence allows attitudes to be extrapolated from adjacent localities when bedded rocks are absent.

Around Kelowna, the Eocene section is approximately 4 km thick and is divisible into three main formations (Fig. 4). The lowest, the Marron Formation, is composed principally of potassic and calc-alkaline volcanics (M. Bardoux and C. Charland, in preparation), and consists of four members, three of which may be correlated stratigraphically with similar units to the south in the White Lake basin (Church 1973, 1982*b*). Age determinations (Table 1) are somewhat scattered, but if the whole-rock determination is ignored, a mean age of about 52 Ma with an uncertainty of about 2 Ma is indicated. Paleomagnetic samples were obtained from 26 sites in the Marron Formation and provided most of the data (Table 2).

Beneath the Marron Formation, there is a lenticular unit consisting of quartz−orthoclase rhyolitic porphyry grading upward into the Trepanier dacite. This unit (sampled at site 21, Table 2, Fig. 3) has a maximum thickness estimated at 500 m, and has yielded a K−Ar date of 50.5 ± 1.0 Ma from unaltered biotite (R. Parrish, unpublished, 1987). Locally, andesitic lavas, petrographically similar to the Shatford Creek Member of Church (1982*a*, 19829b), occur at the bottom of the Marron Formation (Fig. 3). They were sampled at sites 44 and 45 to the east of Okanagan Lake.

Overlying the Trepanier dacite is the Shatford Creek member, followed by a thick sequence of trachytes and trachyandesites that correspond petrographically and geochemically to the Kitley Lake Member of Church (1973). Within the Kitley Lake Member, ash-flow layers occur adjacent to many sites, providing excellent bedding control. Gradationally overlying the Kitley is the Nimpit Lake Member, which consists of trachytes and scoracious trachyandesites. The Kitley Lake and Nimpit Lake members are often hematitic.

On top of the Marron Formation is a 1.2 km thick sequence of interbedded pyroclastic debris flows and lacustrine, fluvial, and alluvial-fan deposits, which are stratigraphically correlated with the White Lake Formation of Church (1973) and which contain palynomorphs of late middle Eocene age (McIntyre, unpublished, 1985). These rocks are tilted about 20° east into the OVF and provide excellent bedding attitudes. At the bottom of the White Lake Formation, thick pyroclastic debris flows carry large cobbles in a fine-grained tuffaceous and mudstone matrix. The cobbles are derived exclusively from the hanging wall. Seven Eocene volcanic cobbles were sampled for a conglomerate test at site 36 (Fig. 3).

The youngest Eocene volcanics are fine-grained dacites, more siliceous and less potassic than the Marron. They occur predominantly east of Kelowna as circular plugs and sills (sites 7, 13, Fig. 3) cutting footwall gneisses, and are probably feeders to outliers of lava of dacitic composition. A K−Ar age of 46 ± 2 Ma (Wm. H. Mathews, unpublished, 1964) has been obtained from biotite separated from fractured dacites (Fig. 4). Taking into consideration this age and their petrography, the dacites are tentatively correlated with the Marama Formation of Church (1973, 1979).

Hornblende gneisses of the footwall of the OVF were also sampled at sites 2, 3, 6, and 14 (Fig. 3). Most gave random magnetization, and demagnetization did not improve groupings. However, some coherent reversed magnetizations were observed at sites 3 and 6, directed nearly parallel to the average mineral-stretching lineation (5° at 110°E), indicating that the remanent magnetization is controlled by the mineral fabric and does not record the paleofield.

Samples have been obtained also from rocks of mid-crustal origin 100 km to the east (Table 2). They are from syenite near Ymir (TSP51) situated in the hanging wall of the easterly dip-

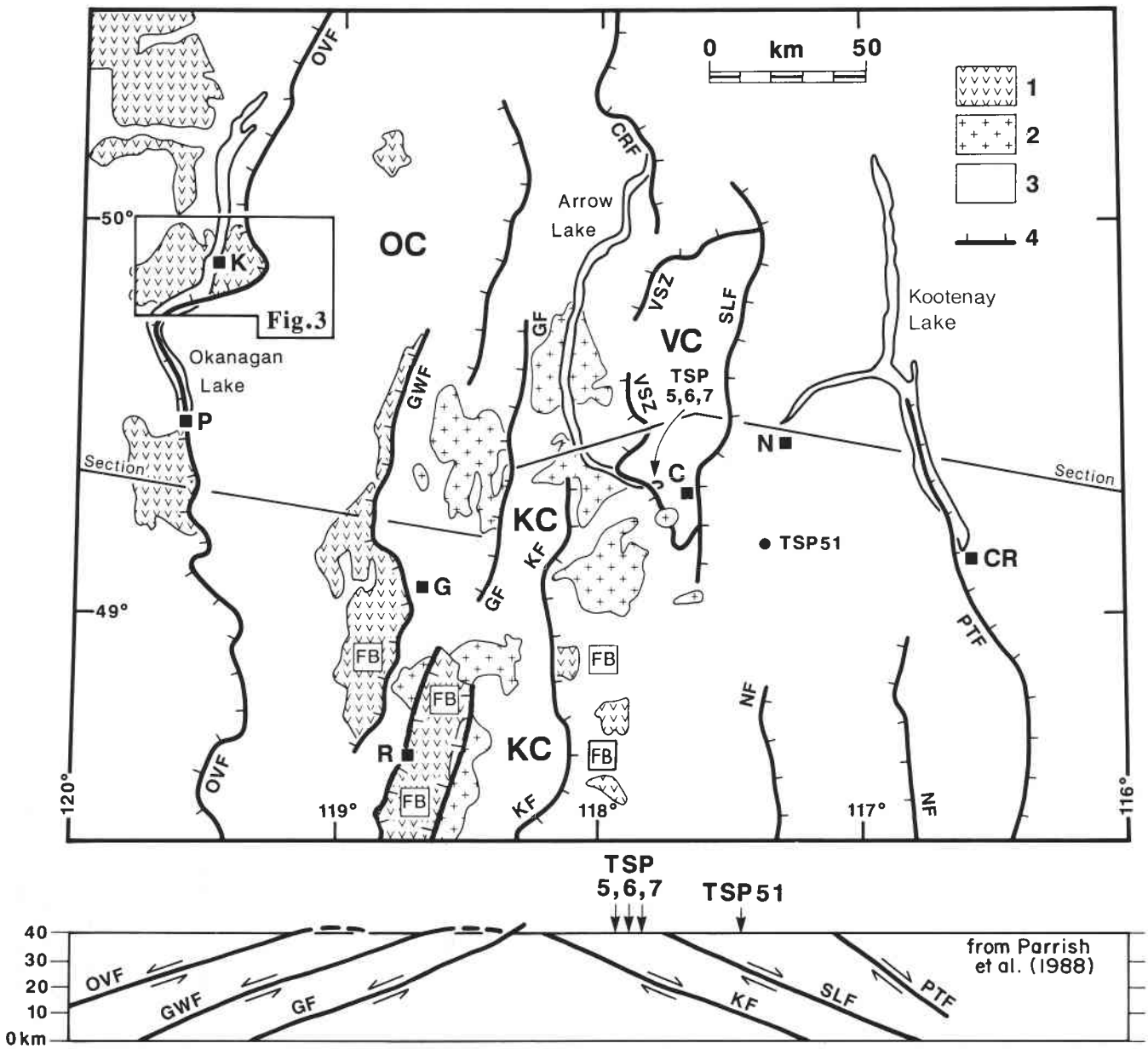


FIG. 2. Zone of Eocene extension, southeastern British Columbia, compiled from Parrish *et al.* (1988). 1, middle Eocene volcanics and volcaniclastic rocks; 2, middle Eocene plutons, syenite, and granite; 3, rocks that predate extension—metamorphic core-complexes, Paleozoic to early Eocene granitoids and metavolcanics; 4, major Eocene low- to moderate-angle extension faults: Okanagan Valley fault (OVF), Slocan Lake fault (SLF), Valkyr shear zone (VSZ), Columbia River fault (CRF), Granby fault (GF), Greenwood fault system (GWF), Kettle River fault (KF), Newport fault (NF), Purcell Trench fault (PTF). Major core-complexes are Kettle – Grand Forks complex (KC), Okanagan complex (OC), Valhalla complex (VC). FB are major sampling areas of Fox and Beck (1985), which also extended 30 km south of the map. C, Castlegar; CR, Creston; FB, sampling regions of Fox and Beck (1985); G, Greenwood; K, Kelowna; N, Nelson; P, Penticton; R, the town of Republic and Republic graben. The hypothetical section below is drawn at a somewhat smaller horizontal scale, but with vertical and horizontal scales the same. Note that SLF is the only fault that has been seismically imaged at great depth (Cook *et al.* 1988). Other faults are extrapolated from surface observations; VSZ is omitted from the section because it predated the Eocene intrusions sampled at sites TSP5–TSP7 and TSP51, data from which are discussed herein (Parrish *et al.* 1988).

ping Slocan Lake fault (SLF) (Carr 1986), and from an altered syenite dyke and metadiorites (sites TSP5–TSP7) along Lower Arrow Lake (Fig. 2). The bodies are late kinematic to postkinetmatic with respect to ductile shearing. Syenite in this vicinity has yielded a U–Pb zircon age of 51.5 ± 0.5 Ma, and metadiorites of TSP5 and TSP6 have yielded 52 Ma U–Pb zircon ages (Parrish *et al.* 1988). Hence, these rocks are essentially coeval with the Marron Formation.

Altogether, a total of about 200 cores were obtained from 41 sites in Eocene rocks. Cores were oriented by magnetic and solar compasses. Generally, two specimens were cut from each core. Tilt corrections were applied by simple rotation about strike. Geological mapping shows that the rocks have been tilted as large panels. Complex folding is absent in the sampling area and plunge corrections are not needed.

Magnetizations

Measurements were made on a Schonstedt spinner mag-

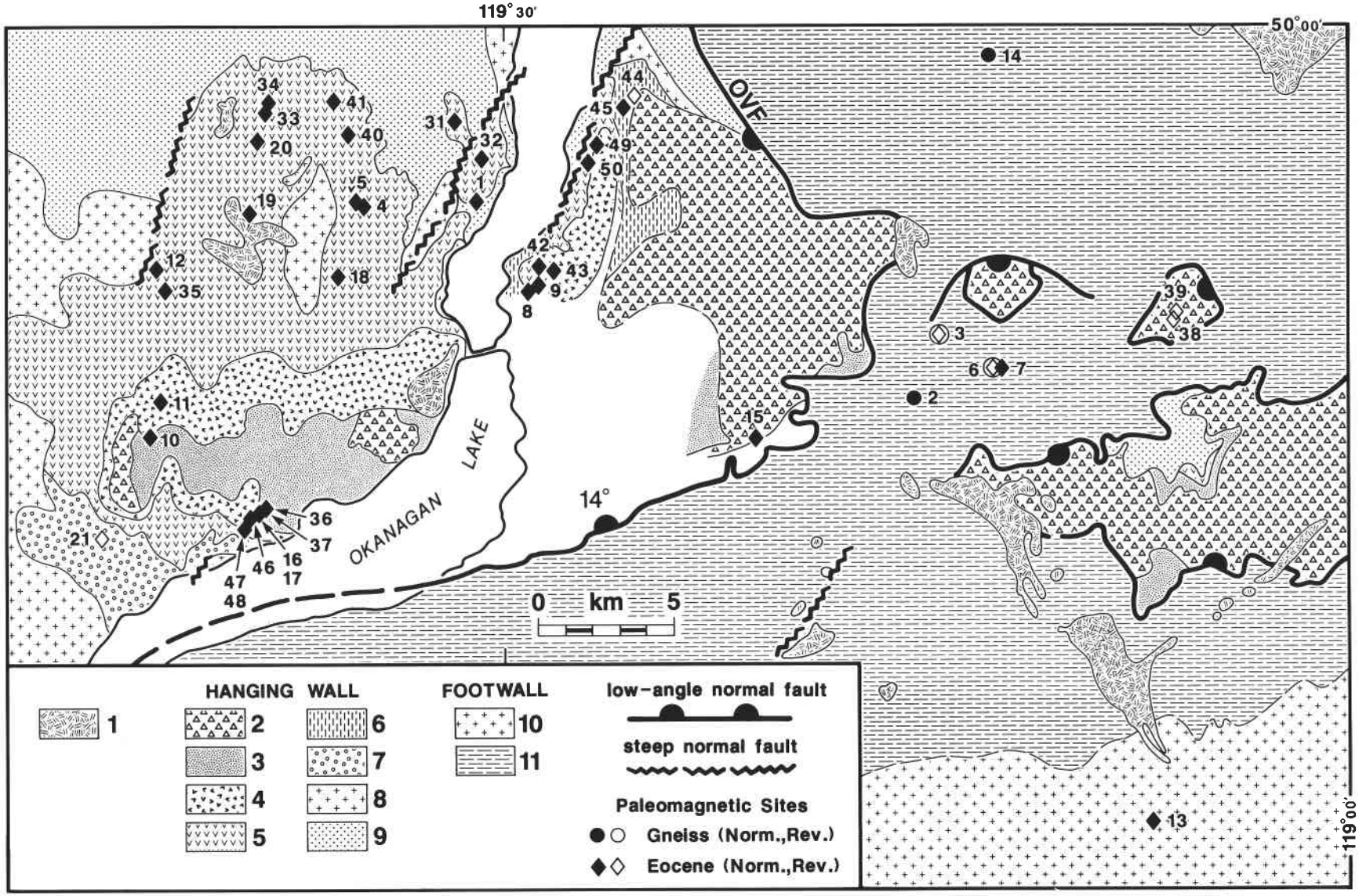


FIG. 3. Geological sketch map with sampling sites. Geology from Little (1961), Church (1973), and Bardoux (1985, 1987). Rock units: 1, Pleistocene basalt and Miocene andesite; 2, Eocene dacites and rhyodacites (? Marama Formation); 3, middle Eocene pyroclastics and lacustrine sediments (White Lake Formation equivalents); 4, Eocene trachyandesites (Nimpit Lake Member); 5, Eocene trachytes (Kitley Lake Member); 6, Eocene andesites (Shatford Creek Member); 7, Quartz–orthoclase porphyry and Trepanier dacite; 8, Mesozoic granodiorites; 9, late Paleozoic metavolcanics; 10 is same as 8 but is deformed; 11, highly strained orthogneiss and paragneiss of the Okanagan dome. The Okanagan Valley fault (OVF) is the main denudation fault marked.

TABLE 1. K–Ar dates from the Marron Formation

Rock unit	Material	Age (Ma)
Kitley Lake Member, Carrot Mountain[a]	Whole rock	44±2
Kitley Lake Member[a]	Biotite	52±6
Kitley Lake Member[a]	Biotite	53
Mafic phonolite, Riddle Creek[a]	Biotite	53±2
Shingle Creek porphyry[a]	Biotite	52±2
Sanpoil volcanics[b]	13 dates	48–53

[a]Church (1978, 1981, 1982*a*).
[b]Marron equivalent in Washington (Pearson and Obradovich 1977).

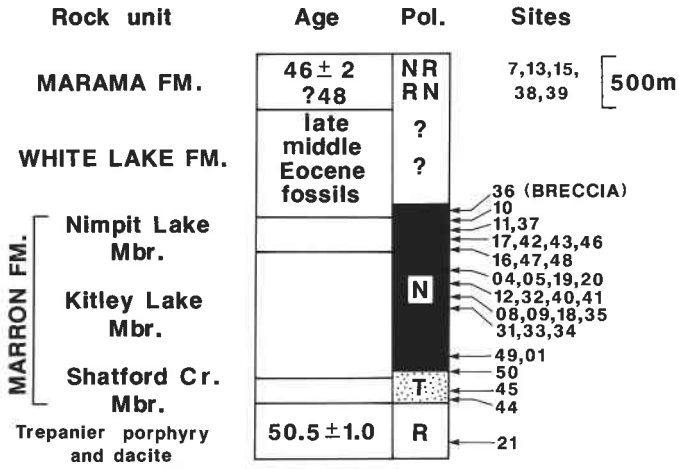


FIG. 4. Magnetostratigraphic column of Eocene volcanic rocks of the Kelowna outlier. Sampling sites numbered as in Fig. 3 and Tables 2 and 3. Stratigraphy after Church (1973) and Bardoux (1987). Radiometric dates from the Marron are given in Table 1. Other dates (all K–Ar) are as follows: 46 Ma Marama, biotite (Medford 1975); 48 Ma Marama (Church 1980*a*), which is queried because it is a whole-rock determination; 50.5 biotite (R. Parrish, unpublished, 1987). N, R, and T denote normal and reversed polarity and transitional directions. The 26 sites in the black sequence all have normal polarity.

netometer (SSM2) controlled by an IBM/PC. Pilot specimens (at least two per site) were demagnetized thermally and by alternating magnetic fields (AF). Stepwise thermal demagnetization was carried out using a Schonstedt TSD-1 furnace, and AF demagnetization, using a Schonstedt GSD-5 tumbling demagnetizer.

The magnetizations are of two types, designated A and B. A-type magnetizations have excellent end points, after removal of small soft magnetizations that are aligned approximately along the present Earth's field and are considered to be recent viscous remanent magnetizations (VRM). Orthogonal plots show regular linear decay to the origin above 300°C and after AF demagnetization in 20 mT and more (Figs. 5, 6). The direction of these high unblocking temperatures, high coercive force magnetizations were obtained by choosing a demagnetization step placed centrally along the line; that is, A-type specimens were cleaned at a single field or temperature. This magnetization is referred to as the "principal" magnetization because it is the largest in magnitude and the most important geophysically.

B-type magnetization, like A-type, is composed of a small VRM and a large principal magnetization, but, in addition, there is commonly a very small magnetization, usually with a very high blocking temperature and very high coercivity. Decay lines on orthogonal plots are linear, but less well-defined than in A-type magnetizations (Fig. 7), and they do not always go precisely through the origin. The principal components of B-type magnetization have been determined by least-squares fitting to linear decay lines using the analysis of Kirschvink (1980) and the computer programme LINEFIND (Kent *et al.* 1983). The small, very high stability magnetizations are attributable to hematite, and probably were produced by incipient recent weathering. Analytical details are given in the footnote to Table 3.

To estimate the time at which magnetizations were acquired a study has been made of a penecontemporaneous volcanic debris flow, and bedding tilt tests have been carried out. Seven petrographically different breccia cobbles were sampled from a debris flow at site 36, all derived from nearby Eocene volcanics. With one exception, the natural remanent magnetizations are directed to the east or northeast and downward (Fig. 8). Clearly, the magnetizations are predominantly overprints, postdating formation of the debris flow, otherwise their directions would have been widely scattered. The exception (specimen 4A) has a principal magnetization with a steep negative inclination that is retained up to 600°C, and that, therefore, predates formation of the debris flow. During thermal demagnetization, the magnetization of the remaining specimens either remained unchanged in direction or migrated in different directions. Some of the "migrating" specimens reach end points (2C of Fig. 8), indicating that the overprint has been removed. Others (8B) do not, indicating incomplete removal. All "migrating" specimens have distributed unblocking temperatures, and all magnetizations removed during demagnetization are directed to the northeast and downwards. Evidently the debris flow was formed at temperatures of about 200°C, which was sufficient to partially remagnetize some specimens and completely remagnetize others. The directions of the high-temperature magnetizations are scattered, indicating that they were acquired before the cobbles were incorporated in the debris flow.

Magnetostratigraphy

Church (1980*b*) has previously found that the natural remanent magnetization (un-demagnetized) of 11 samples spread throughout much of the thickness of the Marron Formation had normal polarity, indicating that only one polarity chron was represented. Although most of our samples are normal, some have reversed polarities. The stratigraphically lowest Eocene volcanic rock in the Kelowna area, the Trepanier porphyry, has reversed polarity (site 21, Table 3, Fig. 4). Samples from the bottom of the overlying Marron Formation (the Shatford Creek Member, sites 44, 45) have directions that are neither normal nor reversed, and differ from one another by much more than would be expected from paleosecular variation. Passing upward into the Kitley Lake Member, the directions stabilize around the normal direction and retain the same polarity through the Marron. Hence, the widely varying directions at the bottom of the Marron Formation, sandwiched as they are between reversed and normal beds, probably represent a reversal transition in the geomagnetic field that occurred at the time the lower Marron lavas were extruded. Because it takes only a few thousand years for the field to reverse (Hospers 1954), this transition could provide a very

TABLE 2. Sampling sites

Site	Rock type	BD/BDDA(RB)	UTM grid reference
M1	Trachyandesite (K)	29/143(*Z*)	319100E, 5534400N
2	Gneiss	16/016	337230E, 5525900N
3	Gneiss	04/220(?)	338600E, 5528100N
M4	Trachyandesite (K)	29/143(*Z*)	314450E, 5533850N
M5	Trachyandesite (K)	29/143(*Z*)	314200E, 5533900N
6	Gneiss	19/036	40900E, 5526400N
m7	Dacite dyke	—	340875E, 5526400N
M8	Trachyandesite (K)	45/075(*X*)	321400E, 5530900N
M9	Green tuff (K)	45/075(*X*)	321550E, 5531050N
M10	Trachyandesite (N)	30/143(*Y*)	305150E, 5524750N
M11	Trachyandesite (N)	30/143(*Y*)	305400E, 5526250N
M12	Trachyandesite (K)	10/315(*Z*)	305500E, 5532300N
m13	Dacite dyke	—	347600E, 5508300N
14	Gneiss	07/028	338800E, 5534650N
m15	Dacite sill complex	13/304(*X*)	331350E, 5524000N
M16	Trachyandesite (N)	55/046(*X*)	308850E, 5521000N
M17	Trachyandesite (N)	55/046(*X*)	308950E, 5521100N
M18	Trachyandesite (K)	10/260(*Z*)	313250E, 5531250N
M19	Trachyandesite (K)	44/315(*Z*)	309550E, 5534050N
M20	Trachyandesite (K)	55/130(*Z*)	309900E, 5537450N
T21	Porphyry	10/090(*Z*)	302700E, 5520750N
M31	Trachyandesite (K)	29/143(*Z*)	318500E, 5537800N
M32	Trachyandesite (K)	29/143(*Z*)	319300E, 5536250N
M33	Trachyandesite (K)	55/130(*X*)	310700E, 5538600N
M34	Trachyandesite (K)	55/130(*Y*)	310650E, 5538850N
M35	Trachyandesite (K)	10/316(*Z*)	305800E, 5531700N
W36	Pyroclastic breccia	28/075(*X*)	309300E, 5521700N
M37	Trachyandesite (N)	55/047(*X*)	309150E, 5521400N
m38	Dacite	—	348750E, 5528300N
m39	Dacite	—	348800E, 5528500N
M40	Trachyandesite (K)	10/261(*Y*)	314100E, 5537350N
M41	Trachyandesite (K)	10/261(*Y*)	313550E, 5538800N
M42	Trachyandesite (N)	45/076(*Y*)	321800E, 5531750N
M43	Trachyandesite (N)	45/076(*Y*)	322300E, 5531650N
M44	Andesite (S)	37/161(*X*)	326300E, 5538800N
M45	Andesite (S)	37/161(*Y*)	325750E, 5537850N
M46	Ash flow (N)	55/046(*X*)	309150E, 5521350N
M47	Lahar (N)	55/046(*X*)	309050E, 5521300N
M48	Lahar (N)	55/046(*X*)	310000E, 5521250N
M49	Trachyandesite (K)	63/104(*X*)	323600E, 5534850N
M50	Ash flow (S)	63/104(*X*)	323500E, 5534400N
TSP51	Unaltered Coryell syenite	—	484900E, 5456800N
TSP5	Pump metadiorite	—	442050E, 5465050N
TSP6	Pump metadiorite	—	442300E, 5465000N
TSP7	Altered syenite dyke	—	442050E, 5465050N

NOTES: (1) M, Marron Formation; W, White Lake Formation; m, Marama Formation equivalent; TSP, Eocene "intrusions"; T, Trepanier porphyry. (2) Rock types are followed by K, Kitley, N, Nimpit, S, Shatford Creek denoting the member to which they belong (Fig. 4). (3) BD and BDDA are the dip of bedding and the bedding down-dip azimuth, both in degrees; RB is the reliability factor for bedding estimates: *X*, very good, *Y*, good, *Z*, obtained by extrapolation from neighbouring sites. (4) Sites TSP5–TSP7 are amphibolite-grade metamorphic rocks.

accurate stratigraphic marker horizon. The dacites we have assigned to the Marama Formation of Church (1973) are the youngest rocks sampled. Two sites are reversely magnetized (38, 39) and three (7, 13, and 15) are normal. Church (1980*b*) observed normal polarity at one site in Marama dacite.

Therefore, at least two reversals occurred, one above and one within the base of the Marron Formation, confining it, in all probability, to one normal polarity chron. Although, the geochronological data are not yet sufficient to assign the normally magnetized Marron to a particular polarity chron, it appears that the 2000 m of lavas were extruded in not more than ~1 Ma, because chrons in the early and middle Eocene do not exceed 1 ma (Harland *et al.* 1982). As already noted, the total accumulation of the Marron volcanics may have been about 2×10^4 km^3, which, if correct, yields an extrusion rate of about 20×10^6 m$^3 \cdot$ a^{-1}, about one-fifth the minimum (100×10^6 m$^3 \cdot$ a^{-1}) estimated for the Columbia River basalts by Baksi and Watkins (1973).

Analysis of data from volcanics

In Table 3, site mean directions are given with respect to

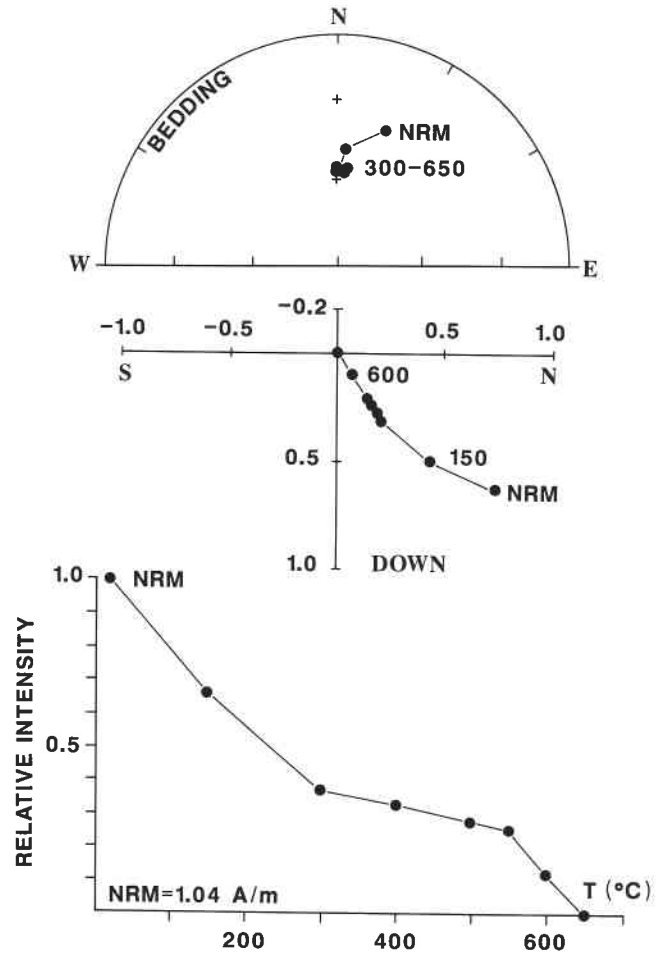


FIG. 5. A1 magnetization. Hematitic lahar, Nimpit Lake Member, site 47. Direction (top) and intensities (bottom) during thermal demagnetization. Orthogonal plot on north–south vertical plane (middle). Inclinations downward.

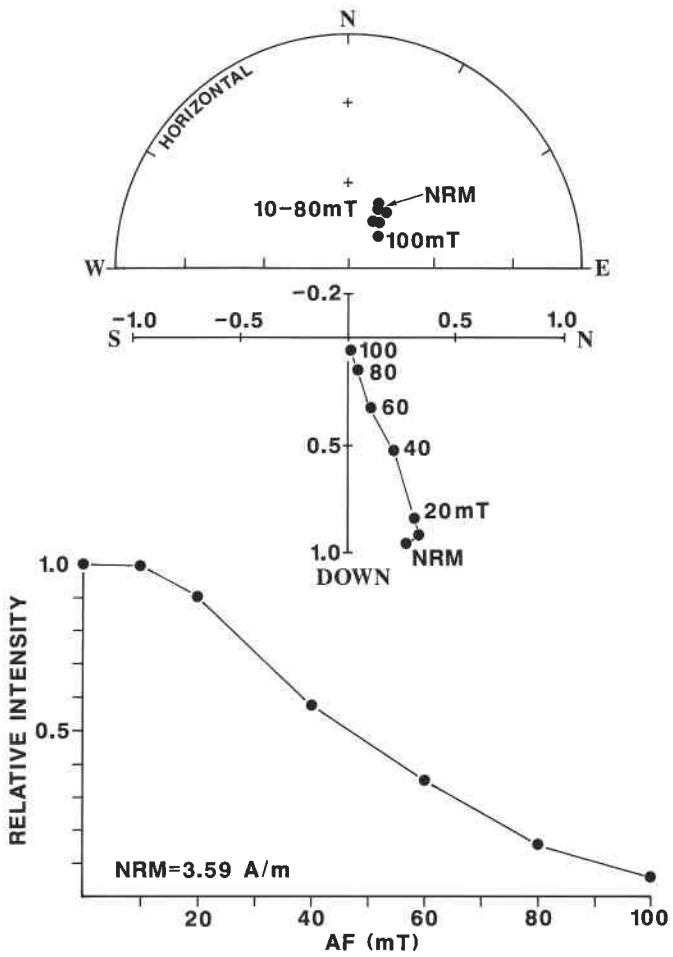


FIG. 6. A2 magnetization. Trachyandesite flow, Kitley Lake Member, site 33. Direction (top) and intensities (bottom) during AF demagnetization. Orthogonal plot on north–south vertical plane (middle). Inclinations downward.

present horizontal plane and to bedding. The exceptions are the dykes at sites 7 and 13, whose directions are given with respect to horizontal only; because no estimate of paleohorizontal can be made, they are excluded from further analysis. The within-site errors (α_{95}) range from 23 to 3°, but generally are between 5 and 9°. Sites 9 and 46 have errors of 20° or more, which is considered too large; they too are not considered further.

The data are summarized in Table 4 and Fig. 9. In all calculations, each site has been give unit weight. In entries (*a*) and (*b*) of Table 4 and Fig. 9, *all* data from the Marron Formation are given with respect to present horizontal and to bedding; tilt corrections do not improve the precision, which remains poor ($k = 8$ before and 9 after correction). Dispersion is much greater than that expected from paleosecular variation. According to the model of McFadden and McElhinny (1984), the precision k expected from a random sample of the geomagnetic field at these latitudes would be much higher—in the range 20–30. The precisions observed from coeval cratonic units are consistent with this model, although the value from the Bearpaw Mountains is greater (Table 5). Certainly, all are much higher than the values ($k = 8$ or 9) obtained from the above analysis of the Marron volcanics. It is possible that paleosecular variation was particularly large during the deposition of the Kelowna volcanics, and that this interval was not represented on the craton. However, if this were so, the bedding tilts observed are such that the precision *in situ* would be less than that after tilt correction, which it is not. This, and the evidence from the debris flow, indicate that some Marron rocks acquired their magnetization before, and others after, tilting.

We begin by adopting the procedure of Schwartz and van der Voo (1984), taking first *all* site directions *in situ*, and then successively applying to them an increasing percentage of the attitudinal correction (Fig. 10). The assumption made is that the rocks acquired their principal magnetization when tilted by the same proportional amount, so that the precision should become maximum when corrections by that amount are made. There is a maximum ($k = 12$) after 60% correction, but the improvement is not great, and the dispersion is still over twice that expected from paleosecular variation at these latitudes. As the tilt correction is increased, the mean direction migrates clockwise, and is closest to the expected Eocene paleofield after 60% correction. The fact that the optimum fit is after 60% correction indicates that about one-half the sites have pretilt, the remainder posttilt, magnetization. This confirms the conclusion reached in the last paragraph.

Having established that magnetizations were acquired over

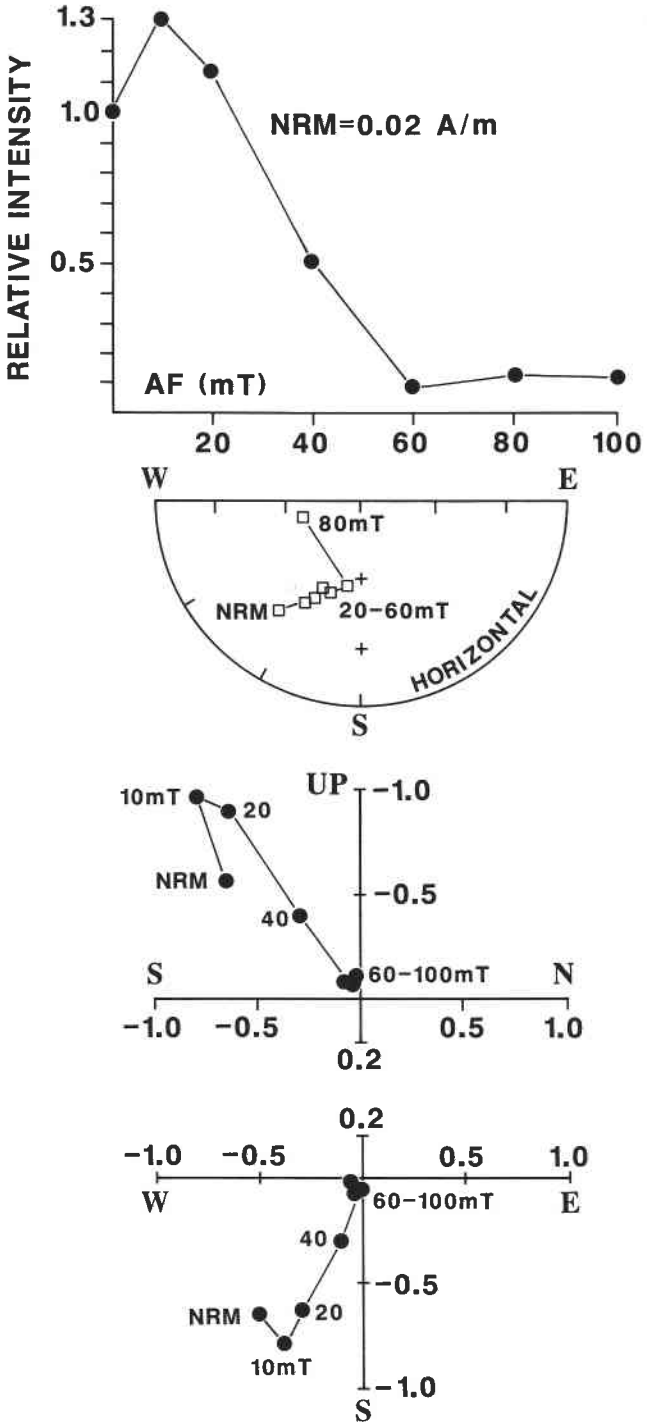


FIG. 7. B2 magnetization. Green ash flow in Shatford Creek Member, site 44. Direction (middle) and intensity (top) changes during AF demagnetization. Orthogonal plots are shown in bottom of figure. Inclinations upward.

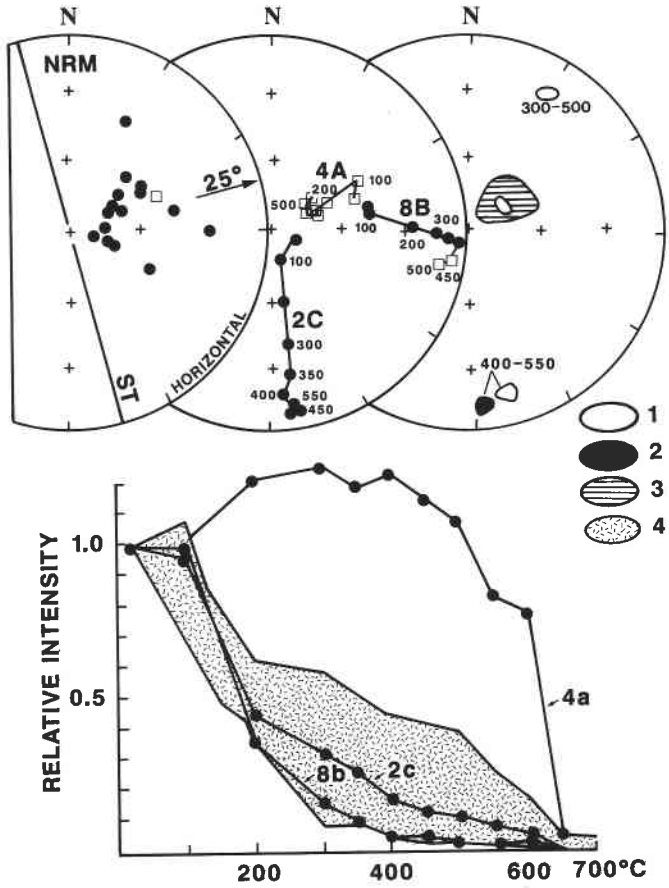


FIG. 8. Debris flow, site 36. The "conglomerate" test is negative for low-temperature magnetizations, positive for high-temperature magnetizations. Solid and shaded symbols denote downward (upward) inclination. Strike and dip of bedding are shown top left. Typical thermal demagnetization curves (bottom) and corresponding direction (middle top) are shown. The shaded area (4) in the lower diagram is the envelope enclosing all thermal demagnetization curves, except that from specimen 4*a*, and shows the distributed nature of unblocking temperatures. (1) and (2) are high-temperature end points in four cobbles, negative and positive inclination respectively. (3) is the envelope of directions of magnetizations (downward inclination) removed in the range 100−350°C.

a substantial time, we proceed by supposing that there are two main intervals of acquisition, one before tilting, and a second, after. Magnetizations were assigned to one interval or the other by comparing their direction, both before and after correction for tilt, with the expected Eocene field direction. In this way two categories of magnetization can be recognized. For example, the principal magnetization at site 47 is, we argue, almost certainly prefolding because the direction, with respect to bedding, is close to the expected Eocene field, whereas the direction with respect to present horizontal implies an improbable 90° anticlockwise rotation (Figs. 5, 11). With respect to bedding, the deviation from the expected Eocene field is small, being that expected either from paleosecular variation or from uncertainties in determining bedding attitude. In contrast, the principal magnetization at site 33 (Figs. 6, 11) is almost certainly posttilting, because the directions with respect to present horizontal are close to the expected Eocene direction, whereas with respect to bedding the directions indicate that site 33 has rotated clockwise by 100° relative to site 47 and was situated about 3000 km further south than site 47; the latter can be discounted because both sites belong to the same mapped volcanic field. Most sites can be readily assigned to one of these two categories. We recognize also category 3 magnetizations, which cannot, because of the low dips (<10°) at the sites from which they were obtained, be allocated to either category. Magnetization type (A or B) and category (1, 2, or 3) are given in Table 3.

TABLE 3. Results by sites

Site	Lat., long. (°N, °W)	C (S)	M (A/m)	D_H, I_H (°)	D_B, I_B (°)	D_C, I_C (°)	k	α_{95} (°)	Type
M1	49.9, 119.5	5 (10)	0.51	346, +62	68, +79	37, +79	36	8	B1
M4	49.9, 119.6	6 (13)	0.40	16, +59	72, +64	20, +61	75	5	A3
M5	49.9, 119.6	6 (12)	1.18	10, +59	69, +67	14, +61	98	4	A3
m7	49.9, 119.2	6 (11)	0.18	357, +65	357, +65	—	63	6	B
M8	49.9, 119.5	7 (13)	0.31	282, +46	358, +71	331, +71	17	10	B1
M9	49.9, 119.5	5 (6)	0.43	28, +64	54, +25	—	17	20	B2
M10	49.8, 119.7	6 (12)	2.07	322, +38	321, +68	321, +62	14	12	A1
M11	49.9, 119.7	6 (12)	1.10	322, +35	321, +65	322, +59	131	4	A1
M12	49.7, 119.9	5 (9)	0.67	48, +70	22, +68	45, +70	121	5	A3
m13	49.7, 119.1	5 (10)	0.17	94, +49	94, +49	—	5	3	B
m15	49.8, 119.3	4 (8)	0.09	321, +68	315, +55	—	28	9	A2
M16	49.8, 119.7	3 (4)	0.60	261, +58	12, +57	358, +66	84	10	B1
M17	49.8, 119.7	4 (4)	0.54	334, +69	24, +26	347, +66	193	7	B2
M18	49.9, 119.6	4 (7)	1.34	43, +56	31, +63	34, +62	90	6	B1
M19	49.9, 119.7	8 (15)	1.18	19, +60	6, +55	17, +60	115	4	A3
M20	50.0, 119.7	4 (8)	0.56	27, +62	95, +36	38, +63	67	7	B2
T21	49.8, 119.7	4 (4)	1.66	125, −51	134, −59	—	41	15	A1
M31	50.0, 119.4	6 (12)	0.51	312, +84	256, +84	311, +61	93	5	A2
M32	50.0, 119.6	4 (5)	2.07	290, +34	270, +56	276, +52	30	14	A1
M33	50.0, 119.6	7 (12)	3.88	24, +72	108, +38	43, +73	83	4	A2
M34	50.0, 119.7	5 (10)	0.74	356, +74	114, +45	16, +77	43	7	A2
M35	49.9, 119.7	6 (10)	1.02	53, +84	349, +79	43, +84	87	5	A3
W36(1)	49.8, 119.7	7 (12)	0.49	41, +70	60, +44	—	147	6	B2
W36(2)		7 (12)	0.49	Random	—	—	2	—	
M37	49.8, 119.7	6 (10)	0.79	257, +27	308, +62	306, +66	34	8	A1
m38	49.9, 119.1	5 (9)	0.01	153, −53	153, −53	—	31	9	B2
m39	49.9, 119.1			Reversed					
M40	50.0, 119.6	6 (12)	0.69	36, +68	12, +74	34, +69	100	4	A3
M41	50.0, 119.6	5 (6)	0.61	15, +66	352, +68	13, +66	172	5	A3
M42	49.9, 119.5	7 (14)	2.91	298, +33	340, +56	328, +54	120	4	A1
M43	49.9, 119.5	9 (14)	1.93	275, +26	301, +66	290, +59	59	5	A1
M44	50.0, 119.4	5 (10)	0.05	202, −51	263, −65	—	73	8	B2
M45	50.0, 119.4	3 (6)	0.16	313, +81	170, +61	—	13	19	BT
M46	49.8, 119.7	6 (7)	0.26	222, −14	220, +41	—	8	23	BT
M47	49.8, 119.7	6 (12)	1.17	272, +54	1, +53	348, +60	178	3	A1
M48	49.8, 119.7	5 (6)	0.27	216, +28	173, +79	—	55	9	AT
M49	50.0, 119.5	7 (12)	0.34	242, +25	187, +52	—	74	5	AT
M50	50.0, 119.5	7 (12)	0.09	312, +36	42, +65	13, +68	76	5	A1
TSP51	49.3, 117.2	5 (10)	0.62	57, +63	—	—	154	4	A?
TSP5	49.3, 117.8	6 (12)	—	68, +51	—	—	14	12	A?
TSP6	49.3, 117.8	5 (9)	—	65, +37	—	—	77	6	A?
TSP7	49.3, 117.8	5 (10)	—	52, +54	—	—	20	11	A?

NOTES: (1) C (S) is the number of cores (specimens). M is the mean intensity of natural remanent magnetization. D_H, I_H is the mean site direction calculated giving specimen unit weight and relative to the present horizontal; D_B, I_B is the same relative to bedding given in Table 1; and D_C, I_C is the same after tilting required to optimize precision. k and α_{95} are precision and radius of the circle of confidence (Fisher 1953). Magnetization type and category are given in the last column. (2) Experimental procedures are as follows: site number [demagnetization applied, as either a single value for A-type magnetization or a demagnetization range for B-type, used for line fitting (number of specimens)]. M1 [15−60 mT (10)]; M4 [80 mT (12), 600°C (1)]; M5 [20 mT (11), 500°C (2)]; m7 [20−80 mT (9), 100−600°C (2)]; M8 [20−60 mT (13)]; M9 [10−60 mT (3), 100−600°C (3)]; M10 [20 mT (11), 600°C (1)]; M11 [20 mT (11), 550°C (1)]; M12 [20 mT (8), 600°C (1)]; m13 [20−80 mT (9), 300−600°C (1)]; m15 [20 mT (9), 550°C (1)]; M16 [20−60 mT (3), 300−700°C (1)]; M17 [40−100 mT (3), 100−650°C (1)]; M18 [20−60 mT (6), 200−700°C (1)]; M19 [15 mT (15), 500°C (1)]; M20 [20−50 mT (8), 100−450°C (1)]; T21 [50 mT (3), 600°C (1)]; M31 [40 mT (11), 500°C (1)]; M32 [40 mT (4), 550°C (1)]; M33 [40 mT (15), 400°C (11)]; M34 [20 mT (9), 400°C (1)]; M35 [20 mT (9), 400°C (1)]; W36(1) [low-temperature component, 100−300°C (7), W36(2) [high-temperature component, 550°C]; M37 [40 mT (5), 400°C (5)]; m38 [10−40 mT (1), 500−600°C (8)]; m39 [general low stability does allow accurate removal of present field VRM, but the most stable part of the remanence is reversed]; M40 [10−60 mT (11), 500−550°C (1)]; M41 [10−80 mT (5), 150−500°C (1), 4 specimens (2 cores) have very high intensities of NRM, their directions diverged from main group, and have sun and magnetic compass orientation that differ by $<10°$—they are regarded as effect of lightning strikes and are rejected]; M42 [40 mT (13), 550°C (1)]; M43 [50 mT (13), 600°C (1)]; M44 [20−60 mT (2), 500−600°C (8)]; M45 [20 mT (2), 10−100 mT (1), 100−400°C (2), 300−650°C (1)]; M46 [20−80 mT (2), 500−600°C (5)]; M47 [40 mT (2), 550°C (10)]; M48 [80 mT (1), 600°C (5)]; M49 [20 mT (8), 550°C (4)]; M50 [10−60 mT (14), 300−500°C (1), 3 specimens (2 cores) were rejected because of high NRM's and 60° divergence from the main group—considered to be effect of lightning]; TSP51 [20 mT (9), 550°C (1)]; TSP5 [40−60 mT (9), 500−600°C (3)]; TSP6 [40−60 mT (4), 550−600°C (5)]; TSP7 [20−40 mT (5), 500−600°C (5)].

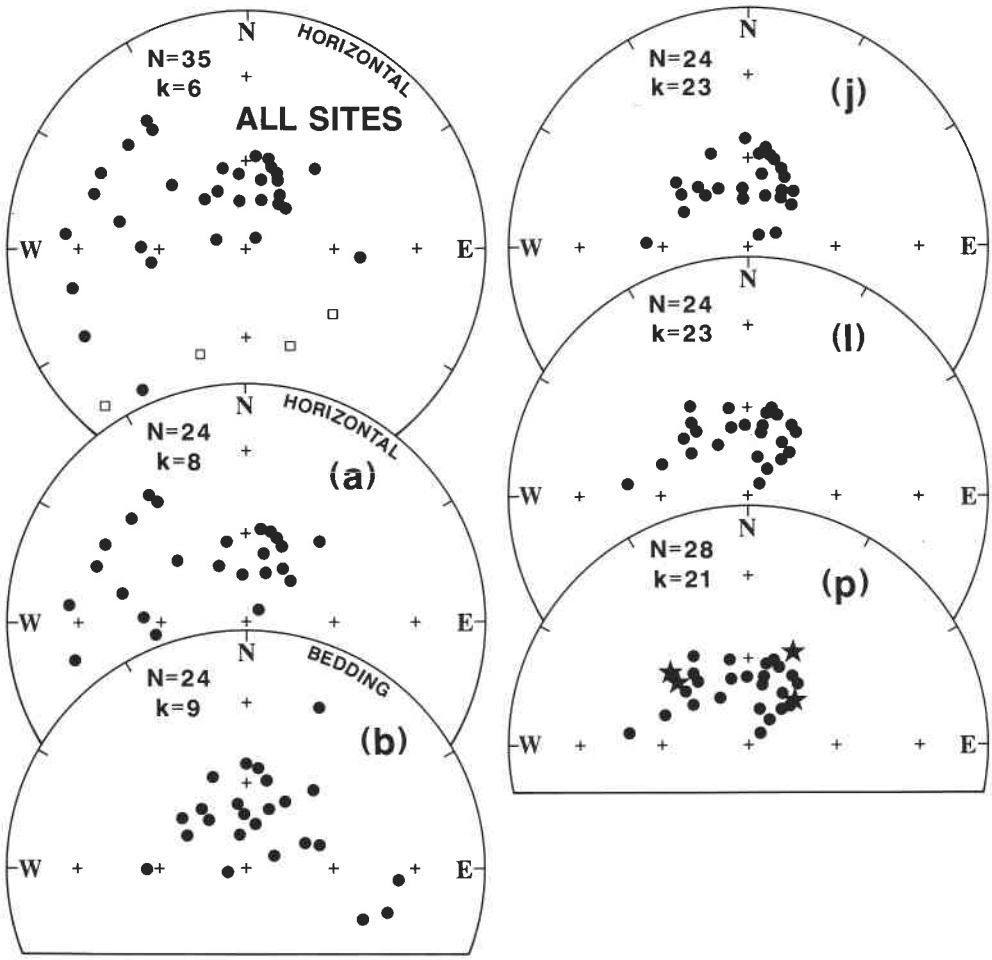


FIG. 9. Mean site directions. Solid (open) symbols denote downward (upward) inclination. Top left shows all sites, including polarity-transition sites. Other stereograms are labelled as in Table 4. Stars in (*p*) are south-seeking directions in reversed sites.

Incremental attitudinal corrections were then applied to the categories separately (Figs. 12, 13; entries (*c*)−(*i*), Table 4). For categories 2 and 3, the assumption that the magnetization is entirely posttilting is probable because the precision (k = 38) is comparable to that expected for paleosecular variations at these latitudes (McFadden and McElhinny 1984) and to those obtained from the craton (Table 5). Even so, partial unfolding yields a small increase after 10% correction (Fig. 13). The mean direction, at first quite close to the expected Eocene paleofield (EE of Fig. 13), migrates away from it as corrections are increased, generating a large and, we argue, erroneous apparent tectonic rotation. For category 1, the precision improves from 9 to 18 after total correction, with a maximum of 22 after 80% unfolding (Fig. 12). As corrections increase, and in contrast to categories 2 and 3, the mean direction migrates towards the expected Eocene paleofield direction, becoming essentially coincident with it. Hence we conclude that category 1 magnetizations are almost, but not entirely, pretilting.

Site directions after optimal tilting (entries *d*), (*g*), Table 4) have been averaged to obtain entries (*l*) and (*m*). A notable feature of Figs. 10, 12, and 13 is the way in which erroneous declination anomalies (apparent tectonic rotations) are generated if incorrect tilt corrections are applied, that is, if incorrect assumptions about the paleomagnetic horizontal are made. To provide an overall summary for Eocene volcanics of the Kelowna outlier (entry (*p*) of Table 4), data from four other sites also have been included. These are from a tuffaceous bed in the Kitley Member (site 9), the Shatford Creek Member (44), the Trepanier porphyry (21), and from the somewhat younger Marama dacite (38).

Data from Eocene "intrusives"

Data from Eocene intrusions of mid-crustal origin in the vicinity of the Slocan Lake fault show excellent internal coherence. An example from the unmetamorphosed syenite near Ymir is shown in Fig. 14. The directions are consistent throughout the demagnetization range, the range of unblocking temperatures and coercive force being typical of those for fine-grained magnetite. The directions are to the northeast and downward, significantly different from those observed in the Kelowna outlier (Table 4).

Tectonics

Time of tilting of Kelowna volcanics

We have just been shown that the magnetization process occurred over a period extending from the time of initial cooling following extrusion until after tilting. If this period had exceeded the length of a polarity chron (less than 1 Ma for the middle Eocene), then the polarity sequence of Fig. 4 would not have been observed. Hence the time lapse between deposi-

TABLE 4. Summary data

Entry	Ref.	N	D, I (°)	k	θ_{63} (°)	α_{63} (°)	α_{95} (°)	Lat., long. (°N, °E)	dm, dp (°)
(*a*) Marron all sites	H	24	323, 64	8	29	7	12	65, 154	18, 15
(*b*) Marron all sites	B	24	020, 72	9	27	6	11	76, 290	19, 16
(*c*) Marron pretilt (category 1)	B	12	343, 69	18	19	6	11	79, 172	18, 15
(*d*) Entry (*c*) 80% tilt correction	B	12	331, 68	22	17	5	10	71, 163	16, 13
(*e*) Marron posttilt (category 2)	H	5	352, 70	31	15	7	14	84, 193	24, 21
(*f*) Marron posttilt (categories 2, 3)	H	12	011, 69	38	13	4	7	82, 306	12, 10
(*g*) Entry (*f*) 10% tilt correction	H	12	016, 70	42	13	4	7	80, 304	12, 10
(*h*) Marron shallow dips (category 3)	H	7	023, 67	58	11	4	8	75, 322	13, 11
(*i*) Marron shallow dips (category 3)	B	7	025, 71	32	14	6	11	74, 301	19, 16
(*j*) Marron (*c*) + (*e*) + (*h*) (model A)	—	24	358, 70	23	17	4	6	86, 218	11, 09
(*k*) Marron (*c*) + (*e*) + (*h*) (model B)	—	24	—	9	26	6	10	86, 228	—
(*l*) Marron (*d*) + (*g*)[a] (model A)	—	24	352, 70	23	17	4	6	83, 195	11, 09
(*m*) Marron (*d*) + (*g*)[a] (model B)	—	24	—	9	27	6	10	96, 230	21, 20
(*n*) Marron (*j*) + extras (model A)	—	28	357, 69	23	17	3	6	87, 204	10, 09
(*o*) Marron (*j*) + extras (model B)	—	28	—	9	27	5	10	87, 217	—
(*p*) Marron (*l*) + extras[b] (model A)	—	28	352, 69	21	18	3	6	84, 184	10, 09
(*q*) Marron (*n*) + extras[b] (model B)	—	28	—	9	27	5	10	85, 197	—
(*r*) Sanpoil volcanics	H	83	027, 66	10	25	3	5	72, 322	08, 07
(*s*) Sanpoil volcanics	B	83	013, 69	12	23	3	5	80, 296	08, 07
(*t*) Eocene "intrusives"	H	4	060, 52	66	10	5	11	43, 329	16, 11

NOTES: (1) N is number of sites; ref. is reference: H, with respect to horizontal, B, with respect to bedding; D, I are declination and inclination of the mean; k is Fisher's precision; θ_{63} is circular standard deviation, α_{63} and α_{95} are errors in mean direction or paleopole at $P = 0.63$ and 0.95; lat., long. are the coordinates of the paleopole; dm, dp are the semiaxes of the paleopole error oval. (2) (*n*), (*o*), (*p*), and (*q*) include site 44 (Shatford Creek Member), site 21 (Trepanier porphyry), site 36(1) (White Lake Fm.), and site 38 (Marama dacite). (3) (*r*) and (*s*) have been calculated from Table 2 of Fox and Beck (1985), excluding sites for which $\alpha_{95} > 10°$; (*r*) is to be compared with entry (*a*) of our analysis, and (*s*) with (*b*). (4) (*t*) Average of data from sites TSP5–TSP7 and TSP51 from Eocene "intrusives."

[a]Considered to be the best summary of Marron Formation.

[b]Considered to be the best summary of Eocene rocks of the Kelowna outlier.

TABLE 5. Paleopoles from cratonic North America

Rock unit	Age (Ma)	N	k	Lat., long. (°N, °E)	A_{95} (°)
Bearpaw Mountains volcanics, Montana[a]	50–54	18	72	81, 198	6
Absaroka Supergroup volcanics, Wyoming[b]	48	19	21	84, 177	10
Highwood Mountains volcanics, Montana[a]	49–53	29	28	81, 167	7
Mean				83, 180	6

NOTES: (1) N is the number of sampling sites; k is the precision of directions. (2) Paleofield direction expected at Kelowna is 352, 68, $\alpha_{95} = 4°$.

[a]Diel *et al.* (1983).

[b]Shive and Pruss (1977).

tion and tilting probably did not exceed 1 Ma; tilting followed deposition very closely.

Absence of rotations

Paleopoles are available from the craton for the interval 53–47 Ma spanning the time within which the Kelowna volcanics lie (Table 5). Other data of about this age are available, some a little older (Paleocene intrusions, Montana, 67–62 Ma, Jacobsen *et al.* 1980), others younger (Rattlesnake Hills intrusions, Montana, 44 Ma, Sheriff and Shive 1980). They have not been included to ensure that comparisons are made within the tightest possible time limits. Data from the Monterey intrusives of Virginia (47 Ma, Lovlie and Opdyke 1974) are not included because they are based on only six sites (see Diehl *et al.* 1983). To have included these data would not have significantly affected the analysis.

The mean cratonic paleopole and that for Eocene volcanics of the Kelowna outlier are in excellent agreement (Fig. 15). This shows that there has been no *large general* rotation of the Kelowna outlier. Motion along the OVF appears to have been simple and extensional, without large systematic rotations about vertical axes, a result which agrees with the local structural evidence reviewed earlier, namely the evidence for simple east-southeast to west-northwest extension of the OVF without rotation about local vertical axes. However, between-site dispersions (Fig. 9(*p*)) are such that small rotations of several degrees among sampling localities are not excluded.

Fox and Beck (1985) have reached the opposite conclusion, based on an extensive study of the Sanpoil volcanics from several grabens spread over 120 km in north-central Washington (FB on Fig. 2). Their rocks are coeval and petrologically similar to those we have sampled. Summaries of their data, relative to bedding and to present horizontal, have been compiled in Table 4 (entries (*r*), (*s*)). The precisions of the Sanpoil

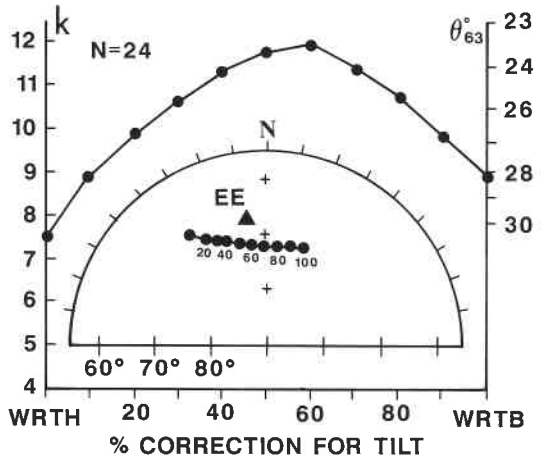


FIG. 10. Variation of precision (k), standard deviation (θ_{63}), and mean direction with degree of unfolding in all sites, Marron Formation. EE is the expected Eocene paleofield direction (Table 5).

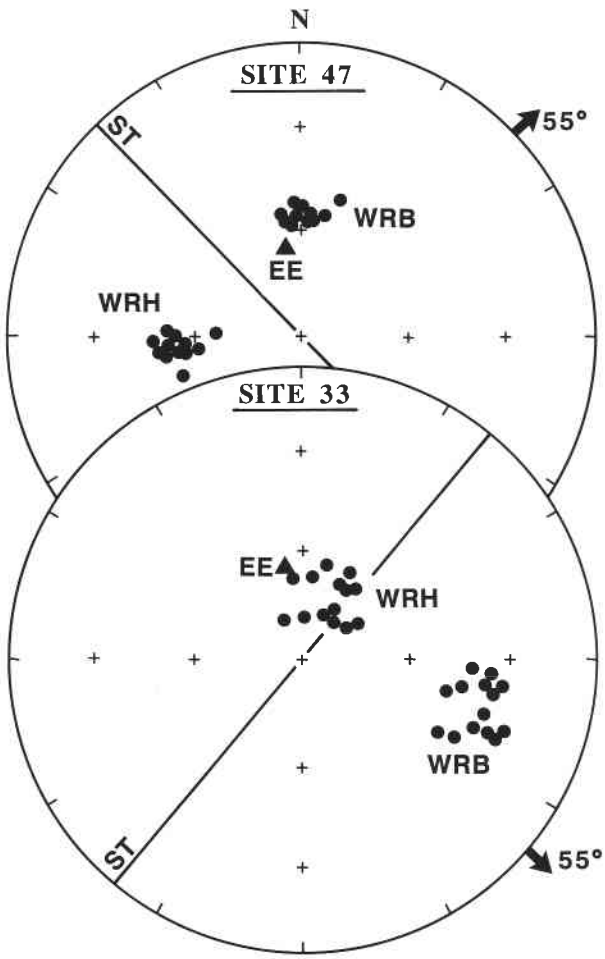


FIG. 11. Pretilting (category 1) (top) and posttilting (category 2) bottom) magnetization. The strikes and tilts are given. WRH and WRB, with respect to horizontal and bedding respectively. EE is the expected Eocene field.

data before and after correction (k = 10 and 12 respectively, Table 4, entries (r) and (s)) are much less than those expected from paleosecular variation, and are comparable to those at Kelowna (k = 8 and 9 respectively, Table 4, entries (a), (b)).

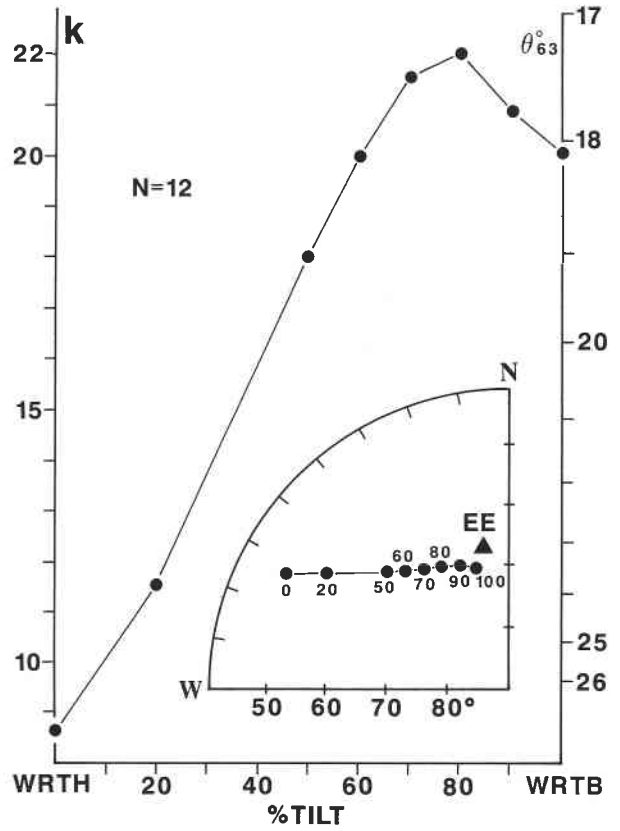


FIG. 12. Variation of precision (k), standard deviation (θ_{63}), and mean direction with degree of unfolding of category 1 sites, Marron Formation. EE is the expected Eocene field.

As in our study, correction for bedding does not improve the precision significantly. Inclinations in the Sanpoil volcanics, like those we have observed, are consistent with cratonic data, and indicate the absence of any significant latitudinal motion. After full correction for bedding tilt at all Sanpoil sites, the declinations are to the north-northeast (Table 4, entry (s)). Fox and Beck (1985) interpreted this as indicating a 25° clockwise rotation of their sampling area.

Likewise, our data, after correction of all sites for full geological tilt (Table 4, entry (b)), would indicate a clockwise rotation of 28°, not significantly different from that invoked by Fox and Beck (1985). We argue that, for the Kelowna volcanics, this would not be a real tectonic effect, but would be an artifact of erroneous bedding corrections made on rocks whose magnetizations were acquired over a time spanning deposition and tilting. When precision is maximized in a manner consistent with the physical evidence, then data from the Kelowna lavas agrees with those from the craton, both in respect to their mean direction and to their precision; when precision is maximized, the data fit very well the cratonic field. The precision obtained is not significantly different, either from that predicted by models of the geomagnetic field over the past few million years, or from that observed in middle Eocene cratonic rocks. Our data are entirely consistent with the null hypothesis that, relative to the craton, there has been no large rotations about a vertical axis in the Kelowna area; there is no basis in the paleomagnetic observations from Kelowna to invoke significant systematic clockwise rotation. We wish to stress that the difference between our interpretation of the Kelowna data and Fox and Beck's does *not* arise

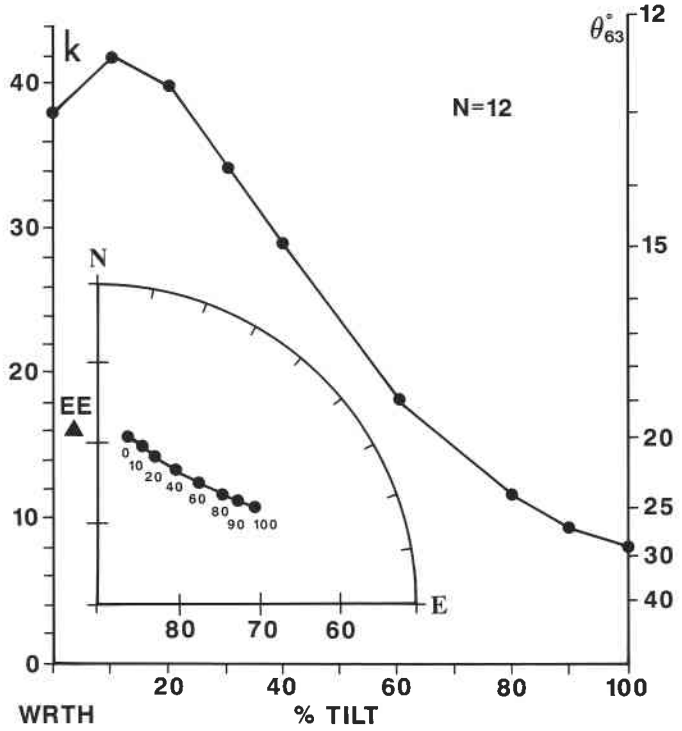


FIG. 13. Variation of precision (k), standard deviation (θ_{63}), and mean direction with degree of unfolding for categories 2 and 3 sites, Marron Formation. EE is the expected Eocene field.

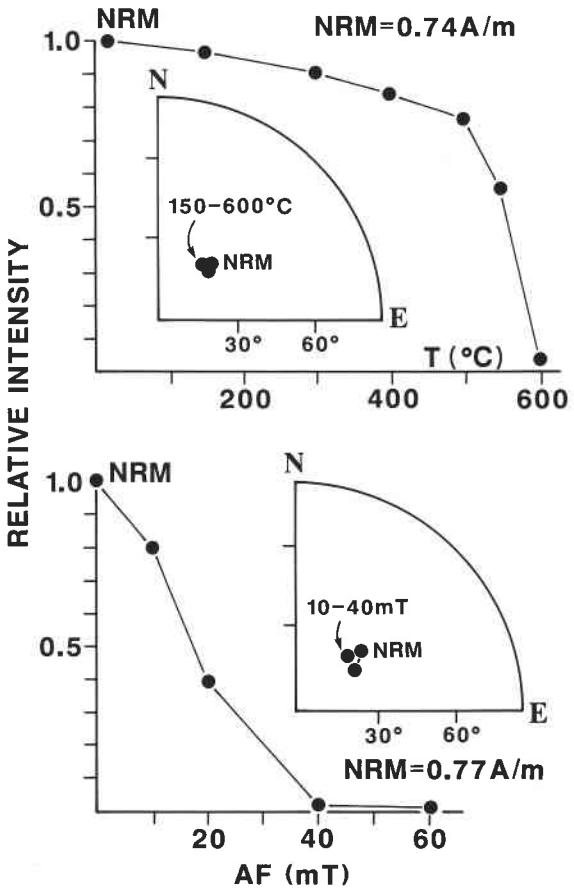


FIG. 14. Thermal and AF demagnetization of Coryell syenite, site 51.

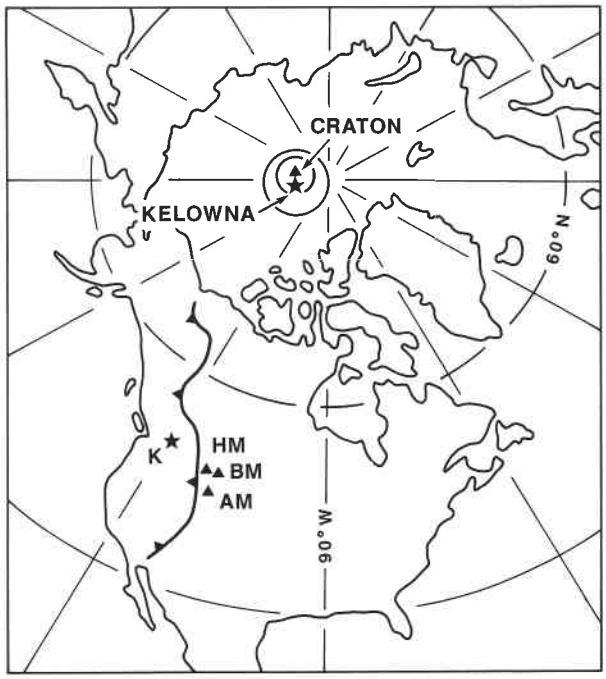


FIG. 15. Paleopoles for middle Eocene rocks of cratonic North America and Kelowna (K) compared. Standard errors ($P = 0.63$) are given. Cratonic data in Table 5.

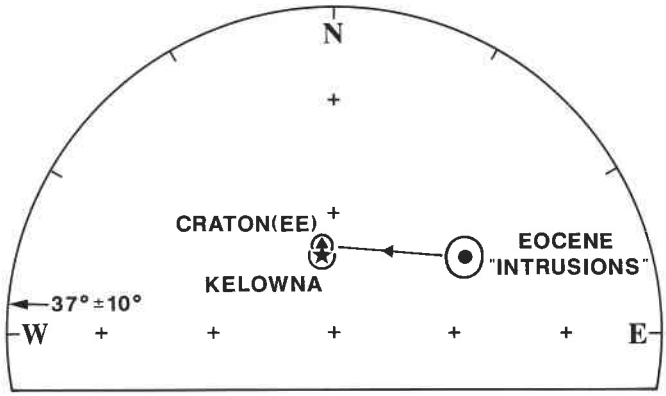


FIG. 16. Mean direction of Eocene "intrusions" from core-complex compared with those expected from craton (EE) and observed at Kelowna. Arrow indicates the tilting proposed and corresponding direction change.

from differences in observations (the data sets seem similar), but from the application of a different mode of analysis that makes different assumptions about the paleomagnetic horizontal.

Tilting in the zone of Eocene extension

Turning now to the discrepancy between data from the Eocene "intrusives" and those from the Kelowna outlier, this discrepancy could have been produced in one of two ways — by tilting of the sampling region to the west (275°) at 37 ± 10° (Fig. 16) or by clockwise rotation of 68 ± 16° combined with translation from the south of 19 ± 12°. The latter can be discounted, because excellent paleolatitudinal conformity has been demonstrated to the immediate west at Kelowna (Fig. 15), to the far west in the Eocene Flores Formation of

western Vancouver Island (E. Irving and M. Brandon, Paleomagnetism of Flores volcanics, in preparation), and to the south in northern Washington (Fox and Beck 1985). Hence a westerly tilt of 37° is the more probable explanation of the aberrant data from the Eocene "intrusives."

In a qualitative sense, the amount of tilt calculated from the paleomagnetic data are reasonable considering that these sites are located between large faults between which "trap door" rotations about horizontal axes may be expected to have occurred (Wernicke 1985). It appears, therefore, that in situations such as this, in which large rotations about vertical axes and translations can be ruled out, paleomagnetic directions can be used to provide quantitative estimates of regional tilt in plutonic and metamorphic rocks of middle and lower crustal origin, rocks which otherwise contain no signature of paleohorizontal. Further studies are in progress to determine whether paleomagnetically observable crustal tilts occur elsewhere in this region.

Possible significance for interpretation of other paleomagnetic aberrancies in the Cordillera

The discordance observed in Eocene "intrusives" is similar to those observed in mid-Cretaceous intrusive rocks of the northern Cascades (Beck and Noson 1972; Irving *et al.* 1985), of the Coast Plutonic Belt (Symons 1977), and of the Intermontane Belt (Armstrong *et al.* 1985). The latter have been interpreted as caused either by systematic tilting over this very wide region of the western Cordillera to the southwest at 30°, or by clockwise rotation of 60° and translation from the south by over 2000 km (Beck *et al.* 1982; Irving *et al.* 1985). This similarity could be a coincidence, and without common cause. On the other hand, the similarity could be taken to indicate a common cause, for example, systematic trapdoor tilting over a much wider area of the western Cordillera in Canada than previously has been contemplated. Although that has never, to our knowledge, been suggested, the fact that the extent of regional extension in southeastern British Columbia has only recently been recognized (Parrish *et al.* 1988) makes it possible that this idea might bear discussion. Such a hypothesis would imply the existence of very large crustal extensional shears in the Intermontane and Coast Plutonic belts, which, to date, have not been recognized.

Acknowledgments

We are grateful for help from J. Baker for magnetic measurements, P. J. Wynne for help in the laboratory and with analysis, R. L. Brown and R. Parrish for discussions and guidance in the field, D. Templemann-Kluit for initiating our collaboration, and M. E. Beck, Jr., B. N. Church, J. K. Park, R. Parrish, and D. T. A. Symons for helpful reviews. Part of the fieldwork of M. B. was carried out when employed as a Geological Survey of Canada field assistant, and part was supported by Natural Sciences and Engineering Research Council of Canada grant A2693 to R. L. Brown.

ARMSTRONG, R. L., MONGER, J. W. H., and IRVING, E. 1985. Age of magnetization of the Axelgold Gabbro, north-central British Columbia. Canadian Journal of Earth Sciences, **22**: 1217–1222.

BAKSI, A. K., and WATKINS, N. D. 1973. Volcanic production rates: comparison of oceanic ridges, islands, and the Columbia River basalts. Science, **180**: 493–496.

BARDOUX, M. 1985. The Kelowna detachment zone, Okanagan Valley, south-central British Columbia. *In* Current research, part A. Geological Survey of Canada, Paper 84-1A, pp. 333–348.

——— 1986. Characteristics of the Okanagan Valley shear zone around Kelowna, south-central British Columbia. Geological Association of Canada, Program with Abstracts, **17**: 43.

——— In press. Geology of the Kelowna area. Geological Survey of Canada.

BECK, M. E., JR., and NOSON, L. 1972. Anomalous paleolatitudes in Cretaceous granitic rocks. Nature (London), **235**: 11–13.

BECK, M. E., JR., BURMESTER, R. F., and SCHOONOVER, R. 1982. Paleomagnetism and tectonics of the Mt. Stuart batholith of Washington: translation or tilt. Earth and Planetary Science Letters, **56**: 336–342.

BROWN, R. L., and JOURNEAY, J. M. 1987. Tectonic denudation of the Shuswap metamorphic terrane of southeastern British Columbia, Geology, **15**: 142–146.

CARR, S. D. 1985. Ductile shearing and brittle faulting in Valhalla gneiss complex, southeastern British Columbia. Geological Survey of Canada, Paper 85-1A, pp. 89–96.

——— 1986. The Valkyr shear zone and the Slocan Lake fault zone: Eocene structures that bound the Valhalla gneiss complex, southeastern British Columbia. M.Sc. thesis, Carleton University, Ottawa, Ont.

CHURCH, B. N. 1973. Geology of the White Lake area. British Columbia Department of Mines and Petroleum Resources, Bulletin 61, pp. 1–120.

——— 1978. Tertiary stratigraphy and resource potential in south-central British Columbia. British Columbia Ministry of Energy, Mines and Petroleum Resources, Geological Fieldwork, Paper 1979-1, pp. 7–15.

——— 1979. Anomalous uranium in the Summerland Caldera. British Columbia Ministry of Mines and Petroleum Resources, Geological Fieldwork, Paper 1980-1, pp. 11–15.

——— 1980*a*. Geology of the Kelowna Tertiary Outlier (west half). British Columbia Ministry of Energy, Mines and Petroleum Resources, Preliminary map 39.

——— 1980*b*. A survey of Cenozoic magnetostratigraphy in south-central British Columbia. British Columbia Ministry of Energy, Mines and Petroleum Resources, Geological Fieldwork 1979, Paper 1980-1, pp. 9–10.

——— 1981. The Riddle Creek uranium–thorium prospect. British Columbia Ministry of Energy, Mines and Petroleum Resources, Geological Fieldwork, Paper 1982-1, pp. 17–22.

——— 1982*a*. Geology and magnetostratigraphy of Miocene basalts of the Okanagan Highlands, British Columbia. British Columbia Ministry of Energy, Mines and Petrolem Resources, Geological Fieldwork, Paper 1983-1, pp. 32–36.

——— 1982*b*. Notes on the Penticton Group, a progress report on a new stratigraphic subdivision of the Tertiary south-central British Columbia. British Columbia Ministry of Energy, Mines, and Petroleum Resources, Geological Fieldwork 1981, Paper 1982-1, pp. 12–16.

COOK, F. A., GREEN, A. G., SIMONY, P. S., PRICE, R. A., PARRISH, R., MILKEREIT, B., GORDY, P. L., BROWN, R. L., COFLIN, K. C., and PATENAUDE, C. 1988. Lithoprobe seismic reflection structure of the southeastern Canadian Cordillera: initial results. Tectonics, **7**: 157–180.

DIEHL, J. F., BECK, M. E., BESKE-DIEHL, S., JACOBSON, D., and HEARN, B. C. 1983. Paleomagnetism of the Late Cretaceous – Early Tertiary north-central Montana alkalic province. Journal of Geophysical Research, **88**: 10 593 – 10 609.

EWING, T. E. 1981. Regional stratigraphy and structural setting of the Kamloops Group, south-central British Columbia. Canadian Journal of Earth Sciences, **18**: 1464–1477.

FOX, K. F., and BECK, M. E. 1985. Paleomagnetic results for Eocene volcanic rocks from northeastern Washington and the Tertiary tectonics of the Pacific Northwest. Tectonics, **4**: 323–341.

HARLAND, W. B., COV, A. V., LLWELLYN, P. G., PICKTON, C. A. G., SMITH, A. G., and WALTERS, R. 1982. A geologic time scale. Cambridge University Press, Cambridge, United Kingdom.

HAUGERUD, R. A. 1986. A hypothesis on the kinematics and thermal

effects of Eocene extension in the Pacific Northwest. Geological Society of America, Abstracts with Programs, **18**: 115.

HELLER, P. L., TABOR, R. W., and SUCZEK, C. A. 1987. Paleogeographic evolution of the United States Pacific Northwest during Paleogene time. Canadian Journal of Earth Sciences, **24**: 1652–1667.

HOSPERS, J. 1954. Reversals of the main geomagnetic field III. Proceedings of the Koninklijke Nederlandse Akademie van Wetenschappen, Series B: Physical Sciences, **57**: 112–121.

IRVING, E., WOODSWORTH, G. J., WYNNE, P. J., and MORRISON, A. 1985. Paleomagnetic evidence for displacement from the south of the Coast Plutonic Complex, British Columbia. Canadian Journal of Earth Sciences, **22**: 584–598.

JACOBSEN, D., BECK, M. E., JR., DIEHL, J. K., and HEARN, B. C., JR. 1980. A Paleocene paleomagnetic pole from North America from alkalic intrusions central Montana. Geophysical Research Letters, **7**: 549–552.

KENT, J. T., BRIDEN, J. C., and MARDIA, K. V. 1983. Linear and planar structure in ordered multivariate data as applied to progressive demagnetization of paleomagnetic remanence. Geophysical Journal Royal Astronomical Society, **75**: 593–621.

KIRSCHVINK, J. L. 1980. The least-squares line and plane and the analysis of paleomagnetic data. Geophysical Journal of the Royal Astronomical Society, **62**: 699–718.

LITTLE, H. W. 1961. Geology, Kettle River (west half), British Columbia. Geological Survey of Canada, Preliminary map 15.

——— 1983. Geology of the Greenwood map area, British Columbia. Geological Survey of Canda, Paper 79-29.

LOVLIE, R., and OPDYKE, N. D. 1974. Rock magnetism and paleomagnetism of some intrusions from Virginia. Journal of Geophysical Research, **79**: 343–349.

MATHEWS, WM. H. 1981. Early Cenozoic resettling of potassium–argon dates and geothermal history of north Okanagan area, British Columbia. Canadian Journal of Earth Sciences, **18**: 1310–1319.

MCFADDEN, P. L., and MCELHINNY, M. W. 1984. A physical model for paleosecular variation. Geophysical Journal of the Royal Astronomical Society, **78**: 809–830.

MEDFORD, G. A. 1975. Geology of the Okanagan Mountain area. Ph.D. thesis, University of British Columbia, Vancouver, B.C.

MILLER, J. D., and KENT, D. V. 1986. Synfolding and prefolding magnetizations in the Upper Devonian Catskill formation of eastern Pennsylvania: implications for the tectonic history of Acadia. Journal of Geophysical Research, **91**: 12 791 – 12 803.

MONGER, J. W. H. 1968. Early Tertiary stratified rocks, Greenwood map-area, British Columbia. Geological Survey of Canada, Paper 67-42.

MONGER, J. W. H., PRICE, R. A., and TEMPELMAN-KLUIT, D. J. 1982. Tectonic accretion and the origin of the two major metamorphic and plutonic welts in the Canadian Cordillera. Geology, **10**: 70–75.

PARKINSON, D. L. 1985. U–Pb geochronology and regional geology of the southern Okangan Valley, British Columbia; the western boundary of a metamorphic core-complex. M.Sc. thesis, University of British Columbia, Vancouver, B.C.

PARRISH, R., and CARR, S. D. 1986. Extensional tectonics of southeastern British Columbia, new data and interpretations. Geological Association of Canada, Program with Abstracts, **11**: 112.

PARRISH, R., CARR, S. D., and PARKINSON, D. L. 1988. Eocene extensional tectonics and geochronology of the southern Omineca Belt, British Columbia and Washington. Tectonics, **7**: 181–212.

PEARSON, R. C., and OBRADOVICH, J. D. 1977. Age and correlation of Eocene rocks in northeast Washington. United States Geological Survey, Bulletin 1433.

ROSS, J. V. 1981. A geodynamic model for some structures within and adjacent to the Okanagan Valley, southern British Columbia. Canadian Journal of Earth Sciences, **18**: 1581–1598.

SCHWARTZ, S. Y., and VAN DER VOO, R. 1984. Paleomagnetic study of thrust sheet rotation during foreland impingement in the Wyoming–Idaho overthrust belt. Journal of Geophysical Research, **89**: 10 077 – 10 086.

SHERIFF, S. D., and SHIVE, P. N. 1980. The Rattlesnake Hills of central Wyoming revisited: further paleomagnetic results. Geophysical Research Letters, **7**: 589–592.

SHIVE, P. N., and PRUSS, E. F. 1977. A paleomagnetic study of basalt flows from the Absaroka Mountains, Wyoming. Journal of Geophysical Research, **82**: 3039–3048.

SYMONS, D. T. A. 1977. Geotectonics of Cretaceous and Eocene plutons in British Columbia: a paleomagnetic fold test. Canadian Journal of Earth Sciences, **14**: 1246–1262.

TEMPELMAN-KLUIT, D., and PARKINSON, D. 1986. Extension across the Eocene Okanagan crustal shear in southern British Columbia. Geology, **14**: 318–321.

WERNICKE, B. 1985. Uniform-sense normal simple shear of the continental lithosphere. Canadian Journal of Earth Sciences, **22**: 108–125.